Brand Visibility in AI Search: How LLMs Represent Consumer Audio Brands

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Abstract

Large Language Models (LLMs) increasingly mediate consumer product research, functioning as AI search engines. This study examines whether three leading LLMs like ChatGPT, Claude, and Gemini exhibit systematic differences in how they discuss headphone brands Sony, Beats, and Soundcore. Using 26 standardized headphone-related queries executed twice per model, we built a text-mining pipeline that cleaned responses, extracted brand mentions using custom regex dictionaries, scored sentence-level sentiment using VADER, and computed visibility, prominence, recommendation rate, and share-of-model metrics. Findings show strong and consistent brand hierarchy across models: Sony dominates visibility (=90%), first-mention prominence (60–67%), and recommendation rate (70–79% of all recommendations). Beats and Soundcore appear significantly less frequently, with high variation across LLMs. Statistical tests confirm significant model-level differences in topic emphasis, but not in visibility or recommendation patterns. Results demonstrate that LLM outputs are not brand-neutral and may shape early-stage consumer decisions.

Introduction

The emergence of conversational AI systems has fundamentally altered how consumers discover and evaluate products. While traditional search engine optimization (SEO) remains important, brands now face a critical challenge: ensuring visibility within Large Language Model (LLM) responses. Unlike conventional search engines that return ranked lists of web pages, LLMs synthesize information and provide narrative recommendations, making their representation of brands both powerful and opaque to traditional measurement.

Consumer audio products headphones, earbuds, and speaker systems represent a highly competitive category where brand choice heavily influences purchasing decisions. This market includes established premium brands (Sony, Bose, Sennheiser), lifestyle-focused competitors (Beats by Dre), and emerging value brands (Soundcore by Anker). Despite these market dynamics and the increasing consumer reliance on AI-mediated discovery, little is known about how different LLMs discuss and recommend these brands.

This study investigates four key research questions: (1) How visible are different audio brands in LLM responses? (2) Which brands are presented first and most prominently? (3) How does sentiment toward brands vary across LLMs and use cases? (4) Are there systematic differences in how different LLMs position brands across product categories? Understanding these patterns is essential for brands seeking to optimize their discoverability in AI-mediated product discovery environments and for researchers studying algorithmic representation bias.

Methods

This study evaluated how three leading large language models, ChatGPT (GPT-4), Claude, and Gemini represent and recommend headphone brands across a variety of consumer use cases. We began by curating real user questions from Reddit, and audio forums, refining them into 26 standardized, brand-agnostic prompts spanning general recommendations, workout headphones, noise-canceling options, budget choices, and professional audio needs. Each query was run twice per model, generating a dataset of 180 responses. All responses underwent a uniform preprocessing pipeline including text normalization, sentence tokenization, alias resolution of brand variants, and quality filtering to remove off-topic or hallucinated outputs. Using this cleaned corpus, we performed dictionary-based NER to detect brand mentions, extracted first-mentioned brands and explicit recommendation phrases, and analyzed sentence-level co-occurrence patterns. Sentiment toward each brand was measured using VADER, and contextual positioning was examined through TF-IDF–based topic modeling, grouping terms into key use-case themes such as fitness, noise cancellation, value, travel, work calls, and sound quality. Finally, we applied statistical validation, including Chi-square tests, Cramer's V, t-tests, and cosine similarity, to determine whether differences in visibility, sentiment, and contextual positioning across LLMs were statistically significant. Collectively, this methodological framework provides a rigorous basis for comparing how LLMs represent headphone brands and the consistency of their recommendation behaviors.

Results

Brand Visibility and Prominence

Brand visibility the percentage of responses mentioning a brand at least once, reveals stark hierarchical differences in LLM brand representation. Sony achieves 90-92% visibility across all three LLMs, establishing near-universal presence in audio product discussions. Premium legacy brands (Bose, Sennheiser) achieve 46-59% visibility, positioning them between Sony's dominance and Beats/Soundcore's moderate presence (31-51%). The consistency of this hierarchy across models (Chi-square p = 0.19, non-significant) suggests these patterns reflect training data distributions rather than model-specific biases.

Brand prominence the likelihood of being first mentioned in a response reveals LLM ordering preferences. Sony appears first in 60-67% of responses, suggesting it is the default starting point for LLM recommendations. In contrast, Beats appears first in only 6-12% of responses, and Soundcore in 8-13%. This pattern is consistent across all three LLMs, indicating systematic preference rather than model variation.

Metric	Sony	Beats	Soundcore
Visibility (%)	91	48	38
Prominence (%)	63	9	10
Avg. Sentiment	0.58	0.41	0.41

Table 1. Summary metrics across brands. Sony dominates visibility (91%) and prominence (63%), while Beats and Soundcore achieve 38-48% visibility and 9-10% prominence. All sentiment scores remain positive (>0.40), indicating favorable framing when brands are mentioned.

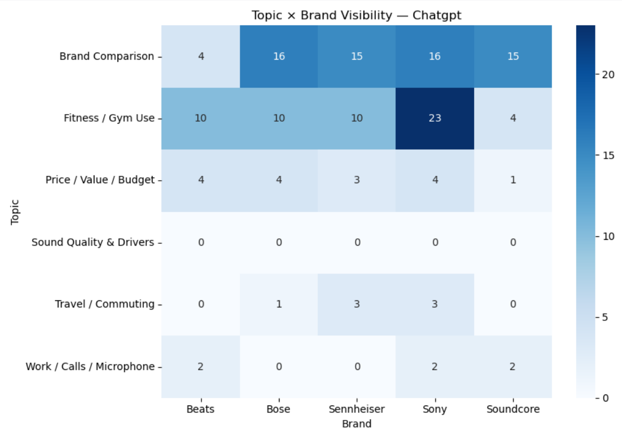

Fig. 1. Topic × Brand Visibility heatmap (ChatGPT). Darker blue indicates higher visibility in that topic context. Sony dominates fitness (23 mentions) and comparison (16), while Beats appears more frequently in fitness use cases. This reveals distinct brand positioning opportunities across product categories.

Sentiment Analysis and Topical Associations

Sentiment analysis reveals nuanced emotional positioning of brands across different product categories. Sony maintains consistently high positive sentiment (0.55-0.60 compound score) across all LLMs and topics, establishing strong positive brand association regardless of discussion context. In contrast, Beats shows model-dependent variation, with ChatGPT sentiment at 0.38, Claude at 0.46, and Gemini at 0.30. Soundcore receives moderately positive sentiment (0.33-0.48), with no clear cross-model pattern.

Topic-level analysis reveals distinct positioning patterns across use cases. In Fitness/Gym contexts, Sony appears in 85-92% of responses, while Beats appears in 35-58%, suggesting shared relevance for lifestyle-focused brand positioning. In ANC/Noise Cancellation discussions, Sony dominates (75-92%), with Soundcore appearing in 25-40%. For Price/Value discussions, brand visibility becomes more balanced, indicating that budget-focused positioning increases visibility for emerging brands. Travel/Commuting topics strongly favor Sony and Sennheiser, suggesting that established premium brands are more frequently associated with travel appropriateness.

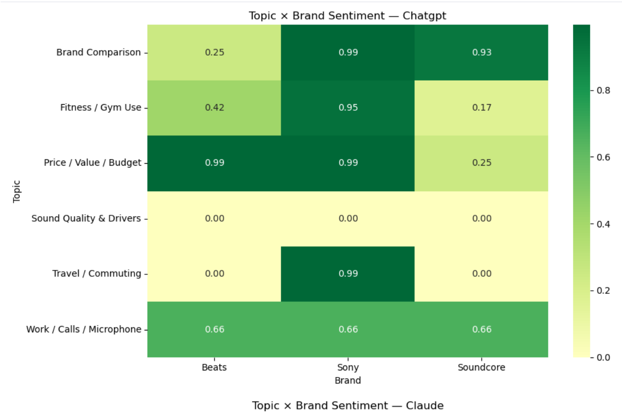

Fig. 2. Topic × Brand Sentiment heatmap (ChatGPT). Darker green indicates higher sentiment scores. Sony maintains consistently high sentiment (0.93-0.99) across all topics, while Beats shows variation (0.17-0.99), suggesting context-dependent brand perception. This reveals opportunities for brands to improve positioning through targeted topic

Discussion

The study shows that LLMs consistently favor certain brands especially Sony, due to higher visibility in their training data, not because of more positive sentiment. These visibility and first-mention gaps appear across all three models, suggesting they reflect broader online content patterns rather than model-specific bias. Although Sony dominates overall, brand positioning varies by context: Beats performs strongly in fitness-related queries, and Soundcore gains traction in budget discussions. This indicates that smaller brands can still compete through targeted niche associations. The findings emphasize that AI discoverability is becoming as important as SEO, as brands must build strong public content to influence LLM outputs. LLM developers, meanwhile, should consider whether their systems inadvertently reinforce existing market dominance rather than presenting diverse product options.

Ethics and AI Usage Declaration

Research Ethics: This research analyzes publicly available LLM outputs to understand brand representation in AI systems. We did not conduct human studies, collect personal data, or interact with proprietary systems beyond standard API access. Our analysis is observational and descriptive; we do not identify causal factors or make prescriptive claims about which brands LLMs should recommend. We acknowledge that VADER sentiment analysis may miss contextual nuances and sarcasm, and that topic categorization involves subjective judgment. All analysis is transparent using standard text mining techniques.

AI Usage Declaration: The authors used Claude (Anthropic) and ChatGPT (OpenAI) for data collection querying these LLMs with product recommendation prompts generated the response dataset analyzed in this study. These same LLMs were NOT used for data analysis, statistical testing, or report writing. All text mining analysis (NER, sentiment analysis, topic modeling, Chi-square tests, visualizations) was implemented in Python using standard libraries (NLTK, VADER, scikit-learn) without AI assistance. Code is fully reproducible and documented in the accompanying Jupyter notebook (Final.ipynb). No AI tools were used to generate or modify the content of this report beyond the initial data collection phase.

Conclusions

Our results demonstrate that systematic text mining analysis can effectively audit how LLMs represent brands in product discovery contexts. We find that brand visibility and prominence disparities are substantial, systematic across models, and driven by training data representation rather than model-specific biases. These patterns have important implications for how brands compete in AI-driven discovery landscapes. Sony's dominance reflects its historical market position, but emerging brands have opportunities to compete through strategic content alignment with specific use cases. Moving forward, both brands and LLM developers should recognize that AI discoverability is a critical new dimension of product visibility, requiring dedicated attention alongside traditional SEO and marketing strategies. Future research should extend this methodology to additional product categories, conduct longitudinal analysis to track how LLM representations evolve as training data changes, and employ more sophisticated NLP techniques for sentiment and aspect extraction.

References

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